FITBUDDY – AI FITNESS PLAN GENERATOR USING GEMINI MODELS

PROJECT DOCUMENTATION

1. INTRODUCTION

FitBuddy is an AI-powered fitness plan generation system designed to provide personalized fitness guidance to users. The system uses Gemini AI models to understand the user's requirements and generate suitable fitness plans. The application helps users create workout routines based on their fitness goals, preferences, available time, and basic personal requirements. It aims to make fitness planning simple, accessible, and easy to understand.

2. PROJECT OBJECTIVES

- To develop an AI-based fitness planning application.
- To generate personalized workout plans.
- To understand user fitness requirements using Gemini models.
- To provide simple and structured fitness recommendations.
- To reduce the difficulty of creating a workout routine manually.
- To provide an easy-to-use interface for users.
- To demonstrate the practical use of Generative AI in fitness applications.

3. PROBLEM STATEMENT

Many people find it difficult to create a suitable workout plan because they may not know which exercises to perform, how to organize their workouts, or how to match exercises with their goals. Traditional fitness planning may require professional guidance or extensive research. FitBuddy addresses this problem by using AI to generate a structured fitness plan based on information provided by the user.

4. EXISTING SYSTEM

- General workout videos.
- Fitness websites and applications.
- Personal trainers.
- Pre-designed workout schedules.
- Manual internet searches.

5. PROPOSED SYSTEM

FitBuddy is proposed as an AI-based fitness plan generator. The user enters their requirements through the application, and the system processes the information using Gemini AI. The generated response can provide a structured fitness plan including workout exercises, sets and repetitions, suggested workout duration, rest periods, and basic general fitness guidance based on the user's stated goal.

6. SYSTEM FEATURES

- User Input – Users can enter basic requirements such as fitness goals, preferred workout type, and available time.
- AI-Based Plan Generation – Gemini models process the user's input and generate a suitable response.
- Personalized Recommendations – The system attempts to tailor the generated plan according to the information supplied by the user.
- Simple Output – The fitness plan is presented in a clear and organized format.
- User-Friendly Interface – The application is designed to be simple to use.

7. TECHNOLOGIES USED

- HTML – Used to create the structure of the web page.
- CSS – Used for styling and designing the interface.
- JavaScript – Used for user interaction and application logic.
- Gemini API / Gemini Models – Used for AI-powered fitness-plan generation.
- Web Browser – Used to access and run the application.

8. SYSTEM REQUIREMENTS

- Hardware: Computer or laptop, minimum 4 GB RAM, keyboard and mouse, internet connection, and modern web browser.
- Software: Windows/Linux/macOS, HTML/CSS/JavaScript development environment, web browser, Gemini API access, and a code editor such as Visual Studio Code.

9. SYSTEM ARCHITECTURE

The basic working architecture of FitBuddy is: User → Web Interface → User Input → Gemini AI Model → Generated Fitness Plan → Output Display. The user provides requirements through the interface. The application sends the relevant prompt to the Gemini model. The model processes the request and generates a response, which is then displayed to the user.

10. WORKING OF THE SYSTEM

- The user opens the FitBuddy application.
- The application displays the fitness input form.

- The user enters their fitness requirements.
- The application collects the entered information.
- The information is converted into an appropriate AI prompt.
- The prompt is sent to the Gemini model.
- Gemini processes the user's requirements.
- The AI generates a structured fitness plan.
- The generated result is received by the application.
- The fitness plan is displayed on the screen.

11. INPUT AND OUTPUT

- Input: Fitness goal, preferred workout type, available workout time, experience level, workout preferences, and other relevant requirements.
- Output: Workout schedule, exercise recommendations, sets and repetitions, workout duration, rest suggestions, and general fitness guidance.

12. ROLE OF GEMINI MODELS

Gemini models are the core AI component of FitBuddy. They are used to understand natural-language user requirements and generate relevant responses. The model can interpret the information provided by the user and produce a structured fitness plan. This demonstrates how Generative AI can be integrated into a web-based application to create personalized content.

13. ADVANTAGES

- Easy to use.
- AI-based fitness planning.
- Personalized responses based on user input.
- Saves time in preparing workout schedules.
- Provides structured information.
- Demonstrates practical Generative AI usage.
- Can be accessed through a web interface.
- Can be enhanced with additional features in the future.

14. LIMITATIONS AND FUTURE ENHANCEMENT

- Limitations: The generated plan depends on the information provided by the user; AI-generated fitness information may not replace professional medical or fitness advice; the application may require an internet connection and API access; the basic version may not track actual workout performance.

- Future enhancements: User login and profile management, workout progress tracking, exercise demonstration videos, fitness history, diet and nutrition guidance, reminder notifications, mobile application support, wearable-device integration, voice-based interaction, and more advanced personalization.

15. CONCLUSION

FitBuddy demonstrates how Generative AI and Gemini models can be used to create a personalized fitness planning application. The system accepts user requirements, processes them using an AI model, and generates a structured fitness plan. The project provides a practical example of integrating AI with web technologies. With further development, FitBuddy can be expanded with progress tracking, mobile support, exercise demonstrations, and other useful fitness-related features.

Note: FitBuddy should be presented as a general fitness-planning project, not as a substitute for a doctor, physiotherapist, or qualified fitness professional.

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A PROJECT REPORT

Submitted in partial fulfilment of the requirements

for the degree of

BACHELOR OF COMPUTER APPLICATION

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ACADEMIC YEAR 2026 – 2027

FITBUDDY: AI-POWERED FITNESS & NUTRITION ENGINE

TECHNICAL SYSTEM SPECIFICATION & ARCHITECTURE REPORT

EXECUTIVE OVERVIEW

FitBuddy is a modern web service engineered to automate dynamic fitness planning through LLM integrations. Built on FastAPI, the core architecture orchestrates asynchronous communication between Google Gemini 2.5 Flash generative models and an underlying SQLite persistence layer via SQLAlchemy ORM.

By evaluating personalized biometrics (age, weight, goal metrics, workout intensity), FitBuddy dynamically constructs full 7-day adaptive exercise regimes, customized feedback-based plan regenerations, and targeted dietary guidance without relying on predefined templates.

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1. PROBLEM DOMAIN & MOTIVATION

Traditional web-based fitness utilities depend heavily on hardcoded static workout databases. These systems fail to accommodate individual progression rate, unique physical profiles, or context-specific adjustments. FitBuddy resolves this bottleneck by implementing an AI-driven feedback loop: combining real-time user metrics with prompt engineering via Google Gemini to render tailored, modular 7-day fitness schedules.

2. ARCHITECTURAL COMPARISON MATRIX

Functional Vector	Legacy Systems	FitBuddy Platform
Core Generation Logic	Pre-written template routines	Generative AI via Gemini Prompting
User Personalization	Grouped into 2-3 broad categories	Individualized per biometric dataset
Iterative Modifications	Requires full manual rewrite	Automated context-aware prompt updates
Data Persistence	Client-side / transient storage	Relational SQLite via SQLAlchemy ORM

3. SYSTEM HARDWARE & STACK PREREQUISITES

Hardware Configuration

- CPU: Modern Dual-Core / Quad-Core Processor x86_64
- System RAM: Minimum 4 GB required
- Storage Space: 1 GB available disk space
- Network Gateway: Continuous Broadband connection for Gemini API calls

Software Stack Specification

- Programming Language: Python 3.10+
- Web API Framework: FastAPI (ASGI execution via Uvicorn)
- Template Engine: Jinja2 Server-side rendering
- ORM & Database: SQLAlchemy with SQLite backend
- Generative AI SDK: Google Generative AI (Model: gemini-2.5-flash)
- Input Validation: Pydantic Data Schemas

4. COMPONENT ARCHITECTURE & DATA PIPELINE

```
Client UI (Jinja2 Templates)
          │
          ▼
HTTP Requests (FastAPI Routing Layer)
          │
```

```
   ┌──────────┴──────────┐
   ▼                     ▼
Pydantic Schemas    AI Engine (Gemini Service)
   │                     │
   ▼                     ▼
SQLAlchemy CRUD  ◄── Plan Response / Dynamic Tips
   │
   ▼
SQLite Storage (fitbuddy.db)
```

Data Processing Sequence

1. User profile submission via frontend web form.

2. Request payload validated by Pydantic; sanitized data passed to backend.

3. AI service formats prompt incorporating user biometrics & fitness targets.

4. Gemini API processes call and generates structured 7-day schedule.

5. Plan record saved in SQLite DB and dynamically rendered on user view.

6. User feedback triggers prompt engineering module to output revised plan.

5. RELATIONAL DATABASE SCHEMA

Database interactions are encapsulated within SQLAlchemy ORM pointing to fitbuddy.db. The relational schema includes three primary tables:

• users: Primary user profile registry (id, name, age, gender, weight, goal, intensity).

• workout_plans: Relational workout history tied via Foreign Key user_id.

• feedbacks: Associated modification requests tied via Foreign Key plan_id.

6. SOURCE FILE HIERARCHY & ROLES

```
fitbuddy_app/
├── main.py            # API routing & endpoint definitions
├── ai_service.py      # Gemini SDK integration & prompt logic
├── crud.py            # Database CRUD operations
├── database.py        # Database connection & session setup
├── models.py          # SQLAlchemy ORM relational models
├── schemas.py         # Pydantic schemas for data validation
├── test_models.py     # Automated unit test suite
├── requirements.txt   # Environment dependency declarations
├── static/            # Frontend CSS, JS, and image assets
└── templates/         # HTML Jinja2 template components
```

7. VERIFICATION & FUNCTIONAL VALIDATION

Case ID	Validation Target	Expected Output	Execution Status
TC-101	User Profile Ingestion	Record persisted in 'users' table	**SUCCESS**
TC-102	7-Day Plan Generation	Structured JSON/text plan from Gemini	**SUCCESS**
TC-103	Plan Persistence	Foreign key mapping in 'workout_plans'	**SUCCESS**
TC-104	Active Plan Retrieval	Latest workout fetched correctly for UI	**SUCCESS**
TC-105	Feedback Input Storage	Feedback string appended to 'feedbacks'	**SUCCESS**
TC-106	Iterative Plan Revision	AI updates plan preserving unaltered context	**SUCCESS**
TC-107	Goal-based Tip Generation	Concise nutrition recommendation produced	**SUCCESS**
TC-108	API Exception Handling	Graceful error trapping on invalid API Key	**SUCCESS**

8. OPERATIONAL STRENGTHS & CONSTRAINTS

Operational Strengths

• High flexibility and adaptability of workout recommendations via LLM inference.

• Interactive feedback loop enabling context-preserving plan revisions.

• Modular separation of web routes, database ORM, and external AI integrations.

System Constraints

• Dependency on low-latency internet connections for Gemini API communication.

• SQLite local database architecture is designed for single-instance or dev deployments.

• General fitness outputs should be verified by certified trainers for medical edge cases.

9. PROJECT CONCLUSION & TECHNICAL REFERENCES

The FitBuddy application effectively showcases the practical fusion of modern Python microframeworks (FastAPI) and Generative Artificial Intelligence (Google Gemini). The clean separation of concerns ensures maintainability and serves as a strong blueprint for scalable web-AI implementations.

Technical References

1. FitBuddy Code Repository (FastAPI, SQLAlchemy, Gemini SDK).

2. Google Generative AI Python API Documentation.

3. FastAPI & Pydantic Core Specifications.

10. APPENDIX: INTERFACE PLACEHOLDERS

[Figure 1.0: User Registration & Profile Intake Module]

[Figure 2.0: AI 7-Day Fitness Plan Output Screen]

[Figure 3.0: Goal-Specific Nutrition & Recovery Tip Output]

[Figure 4.0: Feedback Input & Plan Revision Interface]

[Figure 5.0: Relational Database Schema & File Tree View]